ULTIMATE 2026 AI/ML VAULT

100+ TOP AI & ML INTERVIEW QUESTIONS

Must-know Machine Learning, Deep Learning, LLMs, RAG & AI Agent concepts asked at top tech companies & AI labs.

ML Fundamentals | Deep Learning | NLP & LLMs | AI Agents | MLOps | Ethics & Safety

SWIPE TO MASTER ALL 115 QUESTIONS →

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01 What is the difference between supervised, unsupervised, and reinforcement learning?

02 What is overfitting and underfitting?

03 What is the bias-variance tradeoff?

04 What is regularization? Explain L1 vs L2.

05 What is cross-validation and why is it used?

06 What is the difference between classification and regression?

07 What is a confusion matrix?

08 Explain precision, recall, F1-score, and accuracy.

09 What is ROC-AUC curve?

10 What is the difference between bagging and boosting?

11 Explain Random Forest and how it reduces overfitting.

12 What is Gradient Boosting? How does XGBoost work?

13 What is the curse of dimensionality?

14 What is feature engineering? Why does it matter?

15 What is feature scaling? Normalization vs Standardization?

16 What is one-hot encoding vs label encoding?

17 What is the difference between parametric and non-parametric models?

18 Explain the difference between generative and discriminative models.

19 What is a decision tree, and how does it choose splits?

20 What is entropy and information gain?

21 What is the difference between Gini Impurity and Entropy?

22 What is K-Nearest Neighbors (KNN)? What are its limitations?

23 What is Support Vector Machine (SVM)? Explain the kernel trick.

24 What is Naive Bayes and why is it called "naive"?

25 What is K-Means clustering? How do you choose K?

26 What is the elbow method?

27 What is hierarchical clustering?

28 What is PCA (Principal Component Analysis)?

29 What is the difference between PCA and t-SNE?

30 What is multicollinearity, and how do you detect it?

31 What is A/B testing, and how would you design one?

32 What is class imbalance, and how do you handle it?

33 What is SMOTE?

34 What is the difference between batch, mini-batch, and stochastic gradient descent?

35 What is the learning rate, and what happens if it's too high or too low?

36 What is a loss function? Name a few common ones.

37 What's the difference between L1 loss and L2 loss?

38 What is the difference between a validation set and a test set?

39 What is data leakage, and how do you prevent it?

40 What is hyperparameter tuning? Explain Grid Search vs Random Search vs Bayesian Optimization.

41 What is a neural network, and how does forward propagation work?

42 What is backpropagation?

43 What is the vanishing gradient problem?

44 What is the exploding gradient problem?

45 What is an activation function? Compare ReLU, Sigmoid, and Tanh.

46 Why is ReLU preferred over Sigmoid in hidden layers?

47 What is the dying ReLU problem?

48 What is dropout, and why is it used?

49 What is batch normalization?

50 What is the difference between epoch, batch, and iteration?

51 What is an optimizer? Compare SGD, Adam, and RMSprop.

52 What is weight initialization, and why does it matter?

53 What is a Convolutional Neural Network (CNN)?

54 What is pooling in CNNs? Max pooling vs Average pooling?

55 What is a Recurrent Neural Network (RNN)?

56 What is the vanishing gradient issue in RNNs?

57 What is LSTM, and how does it solve RNN's limitations?

58 What is GRU, and how is it different from LSTM?

59 What is a Transformer architecture?

60 What is self-attention?

61 What is multi-head attention?

62 What is positional encoding, and why is it needed?

63 What is the difference between encoder-only, decoder-only, and encoder-decoder models?

64 What is transfer learning?

65 What is fine-tuning a pretrained model?

66 What is data augmentation?

67 What is a Generative Adversarial Network (GAN)?

68 What is an Autoencoder?

69 What is a Variational Autoencoder (VAE)?

70 What is the difference between GANs and VAEs?

71 What is tokenization?

72 What is the difference between stemming and lemmatization?

73 What are word embeddings? Word2Vec vs GloVe?

74 What is the difference between static and contextual embeddings?

75 What is BERT, and how is it trained?

76 What is masked language modeling?

77 What is GPT, and how does it differ from BERT?

78 What is perplexity in language models?

79 What is beam search vs greedy decoding?

80 What is temperature in text generation?

81 What is top-k and top-p (nucleus) sampling?

82 What is prompt engineering?

83 What is zero-shot, one-shot, and few-shot learning?

84 What is Chain-of-Thought prompting?

85 What is RAG (Retrieval-Augmented Generation)?

86 What is fine-tuning vs prompt engineering vs RAG?

87 What is LoRA and QLoRA?

88 What is RLHF (Reinforcement Learning from Human Feedback)?

89 What is model hallucination, and why does it happen?

90 What is context window, and why does it matter?

91 What is an AI agent?

92 What is the difference between an AI agent and Agentic AI?

93 What is tool calling / function calling in LLMs?

94 What is the ReAct framework?

95 What is a multi-agent system?

96 What is orchestration in agentic workflows?

97 What is memory in AI agents (short-term vs long-term)?

98 What is a vector database, and why is it used in AI systems?

99 What is semantic search vs keyword search?

100 What is chunking, and why does it matter in retrieval systems?

101 What is model drift, and how do you detect it?

102 What is data drift vs concept drift?

103 What is A/B testing in the context of ML model deployment?

104 What is model versioning, and why is it important?

105 What is CI/CD in the context of ML pipelines?

106 What is model monitoring, and what metrics would you track?

107 What is the difference between online and batch inference?

108 What is model quantization?

109 What is model distillation?

110 What is latency vs throughput tradeoff in model serving?

111 What is bias in machine learning models, and how can it arise?

112 What is explainability (XAI) in AI models?

113 What is the difference between interpretability and explainability?

114 What are adversarial attacks in ML?

115 What is data privacy, and how does it relate to model training?

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